HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Prepared detailed **commissioning procedures**, checklists, and test documentation for water and wastewater treatment systems, ensuring alignment with project specifications.
- Verified **mechanical completion** and executed system walkdowns prior to commissioning handover, identifying and documenting **~45 punch-list items** for resolution with subcontractors.
- Executed **cold commissioning activities** including rotation checks, **I/O and loop testing** for over 200 instruments, and valve functionality checks across the WTP and WWTP.
- Managed **hot commissioning** phases, including water testing, biological process start-up and seeding for anaerobic and aerobic systems, and calibration of **chemical dosing systems**.
- Collaborated with automation engineers to test and optimize **PLC and SCADA control strategies**, contributing to the successful integration of new treatment technologies.
- Supported **Site Acceptance Testing (SAT)** and **Operational Qualification (OQ)** protocols for membrane (MBR, UF, RO) and off-gas treatment systems, documenting deviations and resolutions.
- Delivered **operator training** sessions on new equipment and control systems, and assisted in the preparation of **Operations & Maintenance (O&M;) manuals** for final project handover.

SKILLS

Process Engineering – Water Treatment Plant (WTP), Wastewater Treatment Plant (WWTP), Water Recovery Plant (WRP), Drum screening, Dissolved air flotation (DAF), Biological treatment systems, Membrane Bioreactor (MBR), Ultrafiltration (UF), Reverse Osmosis (RO), Activated carbon filtration, Off-gas treatment systems, Chemical dosing systems

Commissioning & Testing – Commissioning procedures, Mechanical completion, System walkdowns, Cold commissioning, Hot commissioning, I/O testing, Loop testing, Instrument functionality, Valve functionality, Alarm testing, Interlock testing, Water testing, Process start-up, Performance testing, Reliability testing, Site Acceptance Testing (SAT), Installation Qualification (IQ), Operational Qualification (OQ)

Automation & Control – PLC, SCADA, Control strategies, P&IDs;

Project Management – Checklists, Test documentation, Punch-list management, Vendor coordination, Subcontractor management, Operator training, O&M; manuals

Core Engineering – Pumps, Blowers, Valves, Root-cause analysis, Troubleshooting, Health & Safety, Permit-to-work, LOTO procedures, Process safety risks

PROJECTS

Industrial Water Treatment Plant Commissioning Support

- Assisted in the execution of commissioning procedures for a large-scale Water Treatment Plant (WTP), focusing on **biological treatment start-up** and membrane system preparation.
- Performed detailed checks on **instrument and valve functionality** across multiple subsystems, identifying and logging **~15 critical issues** prior to hot commissioning phases.
- Supported vendor coordination during the start-up of **chemical dosing systems** and off-gas treatment units, ensuring adherence to safety protocols.

Wastewater Treatment Plant (WWTP) System Integration Testing

- Executed **I/O and loop testing** for the primary treatment stage of a Wastewater Treatment Plant (WWTP), verifying signal integrity for over 100 control points.
- Contributed to **operational handover documentation** by compiling test results and system configuration details for the **MBR and UF membrane** units.
- Participated in troubleshooting sessions for control system anomalies, documenting findings for the automation team and contributing to the resolution of **~8 minor deviations**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)